

Credit scoring using multi-task Siamese neural network for improving prediction performance and stability

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ABSTRACT

A credit scoring model serves as a predictive framework for estimating customer credit risk, specifically the probability of default. The model plays a crucial role in determining the approval of financial transactions, credit limits, and interest rates in financial institutions by predicting a customer's credit risk. Numerous studies have been conducted in the credit scoring field using machine learning techniques. Although the stability of a credit score distribution over time is equally important in credit scoring, most studies have focused solely on improving the model's predictive power. Therefore, this study proposes a multitask learning technique based on Siamese neural networks that simultaneously enhances both predictive power and stability in credit scoring models. Specifically, the proposed model uses personal loan execution data to predict customer defaults while ensuring that the score distribution closely aligns with a predefined golden distribution, thereby securing stability. The golden distribution is a hypothetical five-grade scale derived from scores generated by a pretrained deep neural network. Experimental results show that the proposed model outperforms traditional machine learning and state-of-the-art deep learning models in terms of both predictive power and stability. In particular, the proposed model demonstrates robustness by maintaining high predictive power and stability even in an environment where default rates gently decrease over a long period or where default rates change rapidly over a short period, which can lead to high variability in a model's predictive power and stability.

1. Introduction

A credit scoring model, defined as the use of quantitative models to support decision-making in the consumer credit business, is the backbone of the consumer credit industry. With the rapid growth of this industry, credit scoring has gained considerable economic significance. For example, the balance of consumer credit loans in the United States increased from 4007.0 billion USD in 2018 to 4775.9 billion USD by 2022 (Federal Reserve, 2023). In China, the balance of personal credit loans exceeded 37.8 billion RMB at the end of 2018 (People's Bank of China, 2019). These figures indicate that financial companies need to develop effective credit scoring models for decision-making regarding personal credit loans.

By using credit scoring models to discriminate between risky and non-risky customers (see Fig. 1), financial companies make decisions on credit services such as loan approvals, determining loan amounts, and issuing credit cards (Thomas, 2000). For example, the probability of default (PD) provided by credit scoring determines the interest rate

(Munkhdalai et al., 2019) and affects provisions through the measurement of the expected losses with the total amount of provisions that banks have made (BCBS, 2004). The credit score is calculated using various information such as demographics from loan applications, past performance on debt, income, and possessions (e.g., cars and other residences, if any).

As credit scoring has gained increasing attention, researchers have been actively developing scoring models using various machine learning techniques. Initially, credit scoring was mainly based on traditional models such as logistic regression, decision trees, and simple feed-forward neural networks (Baesens et al., 2003). Moreover, ensemble models, which combine the results of multiple models, have also been actively utilized to improve predictive power (Lessmann et al., 2015; Nalić et al., 2020). Recently, as deep learning models have been successfully applied across diverse industries, researchers have focused on cutting-edge deep learning models for credit scoring as well (Gunnarsson et al., 2021).

Most studies on credit scoring aim to improve the predictive power of

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Fig. 1. Example of credit score cut-off strategy.

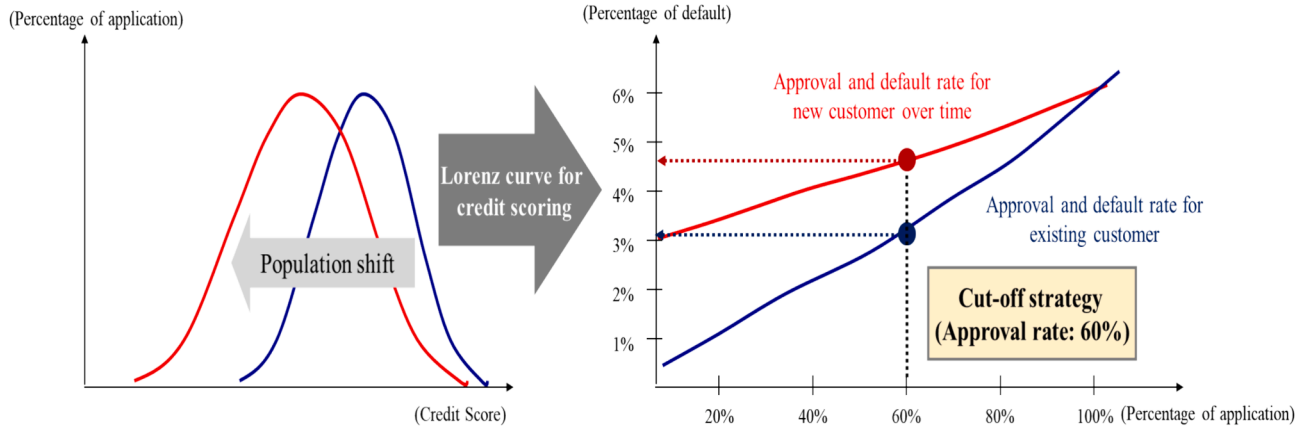


Fig. 2. Lorenz curve for credit scoring.

models. However, the characteristics of financial companies' customers are continually changing owing to various internal and external environmental factors (Brown et al., 2022). Therefore, financial companies must maintain predictive robustness over time. Furthermore, outdated cut-off strategies that do not reflect changes in customer attributes over time could lead financial institutions to approve more high-risk customers, potentially causing losses (see Fig. 2). Approximately 60 % of the applications do not gain the same percentage of default because incoming loan applicants have a higher risk level, leading to a higher default rate. Therefore, financial institutions need to maintain stability in the distribution of customer credit ratings over time to ensure consistency in loan approval criteria (López & Maldonado, 2019).

Additionally, the Basel Committee of the Bank for International Settlements recommends validating internal rating systems and taking appropriate measures (BCBS, 2005). The Basel Committee emphasizes the stability of the credit rating system, because unstable systems demonstrate a significant decrease in predictive accuracy over time. In practice, credit scoring models that follow the internal ratings-based (IRB) approach outlined in the Basel accord must receive approval from financial authorities before being implemented and are subject to periodic supervision (Bank of England, 2022; Monetary Authority of Singapore, 2005). Therefore, maintaining predictive power through periodic retraining can be difficult in an actual financial environment because of the requirement for frequent approval from regulatory authorities. Hence, it is important to develop credit scoring models that can consistently maintain predictive power over time without periodic retraining. To ensure stability over time, out-of-time and out-of-universe validations should be considered for model testing (Markov et al., 2022). Specifically, the Basel Committee recommends examining both the discriminatory power and stability of credit ratings generated by credit scoring models to ensure they adequately differentiate default rates (BCBS, 2023). However, conventional machine learning and deep

learning models have structural limitations in directly enhancing stability. Therefore, most studies have focused on improving customer default predictions.

Given the above discussion, this study proposes a new learning method based on Siamese neural networks to simultaneously achieve high predictive power and stability by introducing the concept of multitask learning. Siamese neural networks are artificial neural networks that convert two sets of input data into embeddings and output the similarity between two embedding vectors (Chopra et al., 2005). In this study, we define predictive power as the binary classification performance in forecasting customer default, and stability as the similarity between the target hypothetical credit score distribution and the default probability distribution trained by the model. In this case, we refer to the hypothetical probability distribution as the golden distribution. This distribution is derived by transforming the scores obtained after training a separate deep neural network (DNN) on customer default data into a hypothetical five-grade scale. To enhance the predictive power and stability, we train the Siamese neural networks to make the difference between the output scores of two customer samples equal to that between the representative scores of the grades in the golden distribution.

We verify the performance of the proposed model using real loan execution data from a P2P lending club in the United States. Beyond testing for predictive power and stability at specific time points, we verify whether the model sustains its performance over time. In the experiments, the proposed model is compared with shallow machine learning and deep learning models at various time points. The results show that our model maintains a higher default predictive power over time and is most effective in maintaining the golden distribution.

The major contributions of this study are as follows:

- We develop a new Siamese network-based multi-task learning technique for credit scoring that not only enhances predictive power but

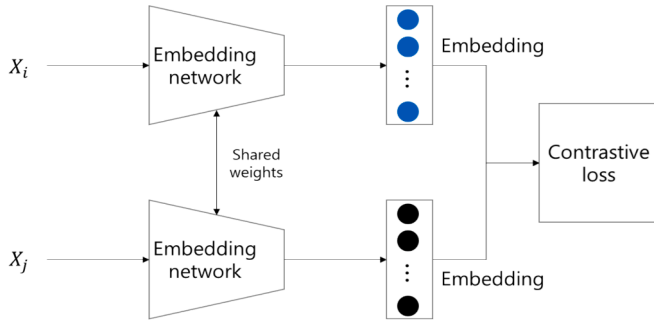


Fig. 3. An overview of the original Siamese network learning.

also improves stability. To the best of our knowledge, this is the first study to develop a credit scoring model that considers both predictive power and stability.

- To ensure model stability, it is essential to maintain a customer's credit grade distribution based on default probability. Therefore, we introduce an additional DNN to create a golden distribution.
- In this study, we define a new loss function for improving default prediction performance and maintaining the golden distribution and propose a new contrastive learning approach based on the defined loss function.
- Experimental results indicate that the proposed model has high predictive power and stability even when customer default rates change over time.

The remainder of this paper is organized as follows. Section 2 briefly describes credit scoring and Siamese neural networks. Section 3 elaborates on the training method of the proposed multitask learning-based Siamese neural network, including the creation of the golden distribution. Section 4 analyzes the comparative experimental results using actual credit data, and Section 5 concludes the paper and suggests future research directions.

2. Related works

2.1. Credit scoring

Credit scoring plays a crucial role in financial institutions' credit evaluation systems by predicting the likelihood of loan repayment. Various machine learning models are used to analyze customer applications and credit information (Thomas et al., 2002). The aim of credit scoring is to accurately classify customers into good and bad risk categories. Let us denote the input vector containing a customer's past loan information as x_i , and the binary variable indicating default as $y^i \in \{0, 1\}$. Here, 0 represents good credit and 1 represents bad credit or default. Credit scoring estimates the probability of default $p(x_i)$ using the trained model and predicts customer default accordingly.

Conventionally, models such as linear discriminant analysis, logistic regression, decision trees, K-nearest neighbors, neural networks, and support vector machines have been used for credit scoring (Henley & Hand, 1996; Huang et al., 2007; Laitinen, 1999; Reichert et al., 1983; Srinivasan & Kim, 1987; West, 2000; Westgaard & Van Der Wijst, 2001). Recent research has primarily focused on improving credit scoring performance by applying advanced machine learning and deep learning technologies. Specifically, researchers have employed ensemble techniques such as bagging or boosting (Wang et al., 2011) and state-of-the-art deep learning models such as convolutional neural networks (CNNs) and long short-term memory to enhance the predictive power of credit scoring models (Wang et al., 2019; Zhu et al., 2018).

As the range of information that can be collected from customers expands, several studies have focused on compensating for the unevaluated parts of existing credit evaluations by utilizing diverse previously

unused information for credit scoring. De Cnudde et al. (2019) propose a model for evaluating the credit of microfinance customers with insufficient credit information using sociodemographic data from Facebook. Djeundje et al. (2021) demonstrate that the performance of credit scoring models can be enhanced using alternative psychological information of customers instead of traditional credit data.

Recently, research has been conducted to develop credit scoring models that maintain robust predictive power over time to establish sustainable loan and credit card evaluation strategies that can withstand external environmental changes. For example, Adams et al. (2010) propose an adaptive linear classifier aimed at improving model performance by considering the changing features of the population over time. Specifically, the authors demonstrate that updating the model periodically with a sliding window approach produces superior performance compared with training the model for the entire period at once. Additionally, Barddal et al. (2020) evaluate the sustained predictive power of models by applying data stream classification techniques to real loan application data that adapt or modify the model in response to changing data distributions. They show that the predictive power of the data stream classification models—adaptive random forest and leveraging bagging—remain stable over time for each month of loan application compared with logistic regression and naive Bayes. Although the study proves that a robust predictive performance can be achieved using data stream classification models, it does not consider the stability of continuously maintaining the distribution of customer credit grades.

2.2. Siamese neural network

Siamese neural networks, as illustrated in Fig. 3, are models that learn the embedding space by considering the class labels of given sample pairs (Chopra et al., 2005; Koch et al., 2015). They are used in various fields, including image search (Wang et al., 2022), image recognition (Koch et al., 2015; Wen et al., 2016), defect detection (Wu et al., 2019; Zhao et al., 2019), and text categorization (Shih et al., 2017). A Siamese network consists of two sub-networks that share weights. When a sample pair x_i and x_j is input, the subnetwork outputs embedding vectors for the pair, and the Euclidean distance d_{ij} between these vectors is calculated. During training, the network learns to minimize the distance between vectors that belong to the same class and maximize the distance between vectors from different classes. The contrastive loss function L_C is defined as

$$L_C(x_i, x_j) = \frac{1}{2} l_{ij} d_{ij}^2 + \frac{1}{2} (1 - l_{ij}) \max(0, \alpha - d_{ij}^2) \quad (1)$$

and a Siamese neural network is trained to minimize it. Here, l_{ij} has a value of 1 if the sample data pair belongs to the same class, and 0 otherwise. α serves as a margin, setting a minimum criterion d_{ij}^2 for maintaining a distance between pairs that belong to different classes.

This study proposes a novel learning method to enhance both the predictive power and stability of credit scoring models by introducing Siamese neural networks. Specifically, we propose a new loss function that considers both predictive power and stability and apply a multi-task learning concept (see Section 3 for details).

3. Proposed model

To develop a credit scoring model that considers both predictive power and stability, we propose the training method shown in Fig. 4. We first define the golden distribution, a target distribution of the proposed model. Because there is no predefined score distribution, we train another DNN separately from the Siamese neural network to predict customer defaults. This DNN calculates the probability of default, which is then converted into the golden distribution. Next, we introduce a multi-task learning-based Siamese neural network to ensure predictive power and stability. Specifically, the model is trained to minimize the

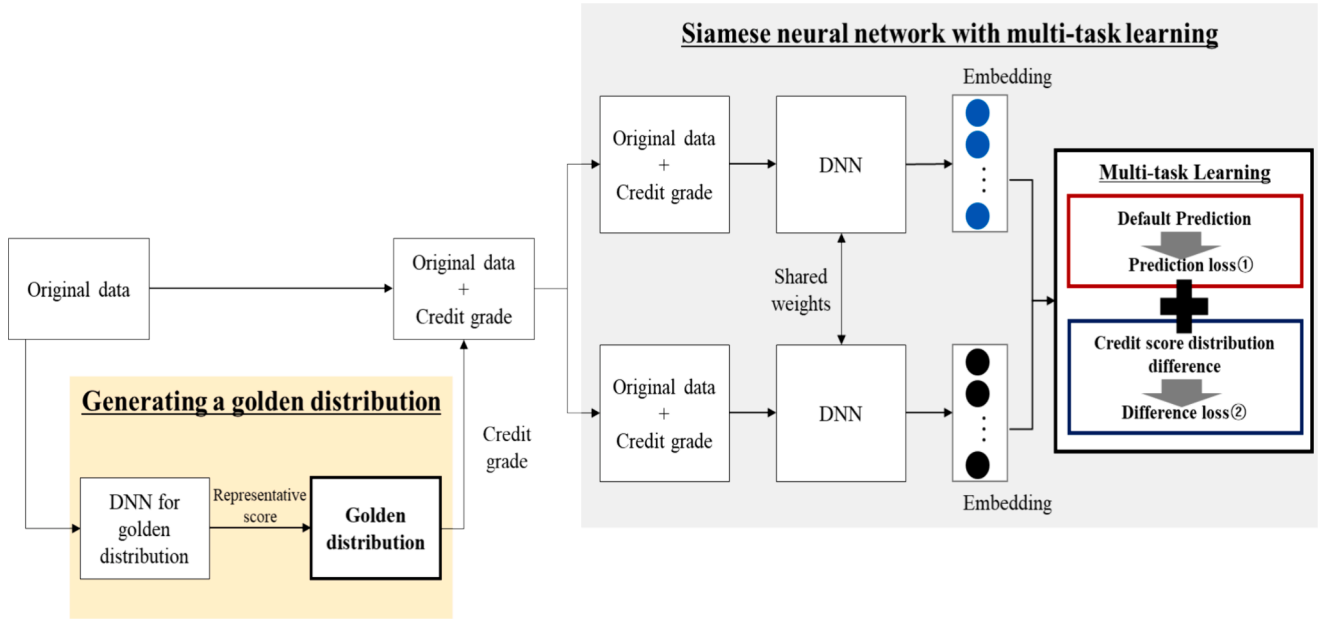


Fig. 4. An overview of the proposed learning method.

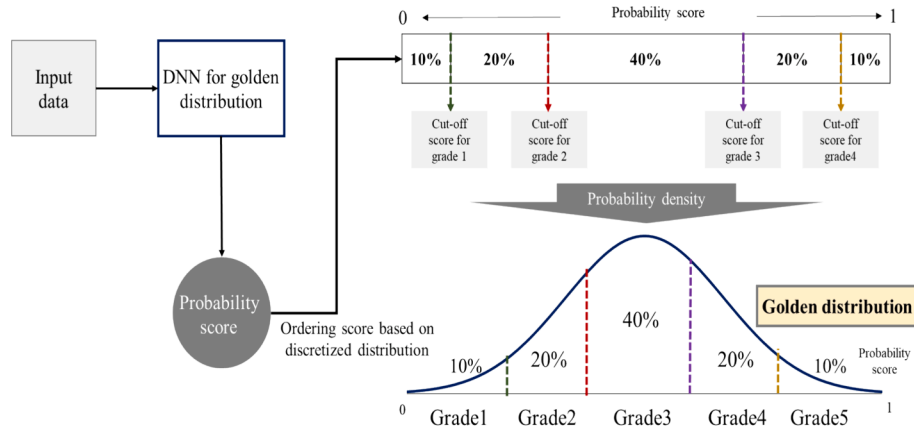


Fig. 5. Golden distribution generation process.

prediction loss for default probability. To maximize stability, we propose a new learning technique that calculates the difference loss by comparing the difference in the estimated scores produced by the two input pairs fed into the Siamese network with the representative grade score difference in the golden distribution. The model is then trained to minimize the difference and prediction losses.

3.1. Golden distribution generation

The golden distribution is a predefined probability distribution used for learning the distribution of default probabilities of input data (see Fig. 5). In this study, we train a DNN to minimize binary cross-entropy, as defined by Equation (2), to generate the golden distribution. The trained DNN outputs the default probability z_i corresponding to the input sample x_i .

$$L_{BCE}(x_i) = -(y_i \log z_i + (1 - y_i) \log(1 - z_i)) \quad (2)$$

where y_i represents the actual default status defined in Section 2.1. To generate the golden distribution, we sort the default probability value for z_i in descending order and categorize it as a credit grade scheme. The golden distribution comprises n credit grades, denoted by G . Each grade

G follows the ratio of P_G . When s_i is the representative score for G in which x_i is located, $s_i = \max(z_i)$.

The calculated default probability is categorized into five credit grades, which we define as the golden distribution. The composition by credit grade is as follows: Grade 1: 10 %, Grade 2: 20 %, Grade 3: 40 %, Grade 4: 20 %, and Grade 5: 10 %. This bell-shaped target distribution is similar to the actual credit scoring distribution in finance. Very few customers have either excellent or very low credit quality, and most borrowers have average credit quality (Zhang & Chi, 2018).

3.2. Siamese neural network training

This study secures high predictive power and stability by incorporating the multi-task learning concept, as shown in Fig. 4. The representative score in Fig. 4 is defined as the maximum score per credit grade in the golden distribution. Specifically, two randomly selected input samples are fed into a Siamese network to generate embedding vectors. The relationship between the two vectors is then learned via contrastive learning. First, customer information, including credit history and outstanding loans, is used as input to predict the probability of default. Here, the loss function employed is binary cross-entropy.

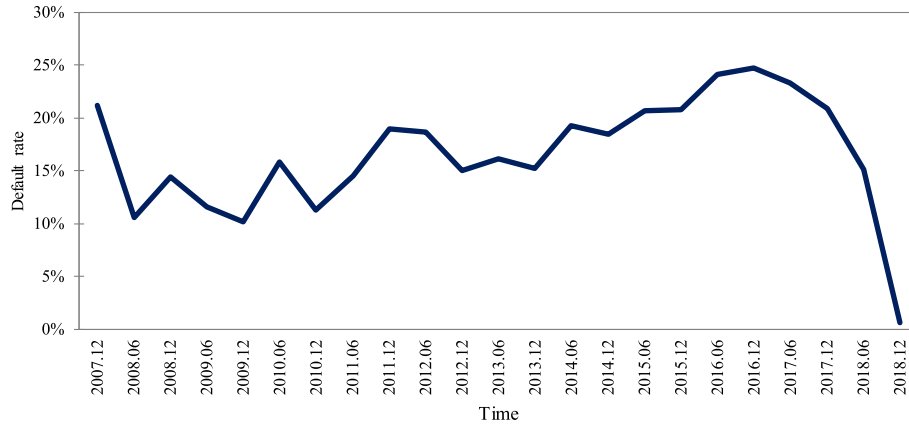


Fig. 6. Lending club default rate over time.

Table 1

Summary of test periods.

Experiment no	Period	Training		Test		Difference (A-B)
		Period	Default rate (A)	Period	Default rate (A)	
1(Long term)	Period 1	'16.1 ~ '16.12	23.7 %	'17.1 ~ '17.3	22.8 %	0.9 %
	Period 2	'16.4 ~ '17.3	24.4 %	'17.4 ~ '17.6	23.6 %	0.8 %
	Period 3	'16.7 ~ '17.6	23.9 %	'17.7 ~ '17.9	23.6 %	0.3 %
	Period 4	'16.10 ~ '17.9	23.4 %	'17.10 ~ '17.12	21.1 %	2.3 %
	Period 5	'17.1 ~ '17.12	22.8 %	'18.1 ~ '18.3	18.9 %	3.9 %
	Period 6	'17.4 ~ '18.3	21.8 %	'18.4 ~ '18.6	17.1 %	4.7 %
2(Short term)	Period 1	'17.9 ~ '18.5	20.1 %	'18.6 ~ '18.8	11.4 %	8.7 %
	Period 2	'16.1 ~ '16.3	20.3 %	'16.4 ~ '16.6	25.2 %	-4.9 %

Table 2

Default prediction models used for comparison.

Classification	Comparison Models
Shallow Machine Learning Models	Logistic regression, decision tree, k-nearest neighbors (KNN), support vector machine (SVM)
Ensemble Models	Random forest, AdaBoost, GradientBoost, XGBoost
Deep Learning Models	MLP, CNN, FCN, ResNet, Inception, NN-RF, RF-NN
Data Stream Classification Models	Adaptive random forest, leveraging bagging
Customized Model	Multi-task NN

$$L_{BCE}(x_i) = -(y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)), \quad (3)$$

where $\hat{y}_i$ represents the predicted default probability for x_i .

Second, a new form of contrastive learning technique is proposed to maintain the golden distribution of scores based on credit grades. Specifically, the output scores of the two input samples are checked against the golden distribution to ascertain their corresponding credit grades. The learning process then ensures that the estimated score difference aligns with the grade difference in the golden distribution. This learning approach indirectly ensures that the overall distribution of the estimated score closely resembles the golden distribution by ensuring that the score difference between the two samples follows a representative score difference according to the grade within the golden distribution. To achieve this, we use the mean squared error between the grade score difference in the golden distribution and the output score difference of the input samples as the learning loss, as defined by

$$L_{MSE}(x_i, x_j) = \left((\hat{y}_i - \hat{y}_j) - (s_i - s_j) \right)^2 \quad (4)$$

Let us assume that the representative score corresponding to each credit grade in the golden distribution is defined as follows: (Grade 1) s_1 : 0.1, (Grade 2) s_2 : 0.3, (Grade 3) s_3 : 0.7, (Grade 4) s_4 : 0.9, (Grade 5) s_5 : 1.0. Let us also assume that x_1 and x_2 belong to Grade 2 and Grade 3,

Table 3

Hyper-parameter setting in comparison models.

Comparison models	Hyper-parameter setting
Logistic regression	penalty: l2
Decision tree	criterion: gini, splitter: best, min_samples_split: 2
KNN	n_neighbors: 5, weights: uniform, algorithm: auto, leaf_size: 30
SVM	penalty: l2, loss: squared_hinge
Random forest	n_estimator: 100, criterion: gini, max_depth: 7
AdaBoost	n_estimators: 50, learning_rate: 1.0, algorithm: SAMMER
GradientBoost	loss: log_loss, learning_rate: 0.1, n_estimators: 100, min_samples_split: 2
XGBoost	booster: gbtrees, max_depth: 6, eta: 0.3
MLP	hidden_layer_size: 200, activation: ReLU, solver: Adam, learning_rate: 0.01
CNN	layer_size: 6, activation: sigmoid, padding: same, kernel_size: 7
FCN	layer_size: 11, activation: ReLU, padding: same
ResNet	activation: ReLU, padding: same, solver: Adam
Inception	activation: ReLU, padding: same, solver: Adam
NN-RF / RF-NN	n_estimator: 100, max_depth: 7, activation: ReLU
Adaptive random forests	n_models: 10, max_features: sqrt
Leveraging bagging	n_models: 10, bagging_method: bag
Multi-task NN	hidden_layer_size: 3, activation: ReLU

respectively. By Equation (4), when x_1 and x_2 are fed during training, the Siamese neural network is trained to map the difference between $\hat{y}_1$ and $\hat{y}_2$ to 0.4, which is the representative score difference between Grade 2 and Grade 3. This learning process helps maintain the distribution of score differences between samples according to the golden distribution and ultimately improves the stability of the model.

Ultimately, we define the total loss by summing Equations (3) and (4) as

$$L_{Total}(x_i, x_j) = \alpha(L_{BCE}(x_i) + L_{BCE}(x_j)) + (1 - \alpha)L_{MSE}(x_i, x_j) \quad (5)$$

Table 4

Results of the golden distribution from P2P Lending Club data for a long-term period.

Credit Grade	'16.1 ~ '16.12			'16.4 ~ '17.3			'16.7 ~ '17.6		
	Number of customers	Composition ratio	Score	Number of customers	Composition ratio	Score	Number of customers	Composition ratio	Score
Grade 1	29,258	10.0 %	0.01	22,755	10.0 %	0.03	20,910	10.0 %	0.02
Grade 2	58,515	20.0 %	0.11	45,511	20.0 %	0.11	41,819	20.0 %	0.13
Grade 3	117,032	40.0 %	0.34	91,022	40.0 %	0.29	83,638	40.0 %	0.30
Grade 4	58,515	20.0 %	0.48	45,511	20.0 %	0.52	41,819	20.0 %	0.44
Grade 5	29,258	10.0 %	0.93	22,755	10.0 %	0.92	20,910	10.0 %	0.92
CreditGrade	'16.10 ~ '17.9			'17.1 ~ '17.12			'17.4 ~ '18.3		
	Number of customers	Composition ratio	Score	Number of customers	Composition ratio	Score	Number of customers	Composition ratio	Score
Grade 1	19,199	10.0%	0.01	16,870	10.0%	0.02	14,431	10.0%	0.01
Grade 2	38,397	20.0%	0.10	33,740	20.0%	0.12	28,864	20.0%	0.08
Grade 3	76,796	40.0%	0.28	67,480	40.0%	0.30	57,726	40.0%	0.28
Grade 4	38,397	20.0%	0.45	33,740	20.0%	0.47	28,863	20.0%	0.44
Grade 5	19,199	10.0%	0.91	16,870	10.0%	0.94	14,431	10.0%	0.94

Table 5

Results of the average composition ratio between training and test period.

Credit Grade	Training(A)	Test(B)	Difference(B-A)
Grade 1	10.0 %	3.8 %	-6.2 %
Grade 2	20.0 %	24.0 %	4.0 %
Grade 3	40.0 %	43.4 %	3.4 %
Grade 4	20.0 %	20.1 %	0.1 %
Grade 5	10.0 %	8.7 %	-1.3 %

where α is a balancing parameter. Finally, the proposed learning algorithm is summarized as Algorithm 1. Parameter θ is updated using stochastic gradient descent.

Algorithm 1: Pseudo code for our proposed contrastive learning algorithm
Input: Training dataset $D = \{(x_1, y_1), \dots, (x_N, y_N)\}$, balancing parameter α
1: **Produce** default probability $z_i = p(y_i = 1 | x_i, \theta_G)$ for all $x_i \in D$ using DNN for golden distribution (with parameter set θ_G) trained on D
2: **Assign** credit grade according to $p(y_i = 1 | x_i, \theta_G)$ for all $x_i \in D$
3: **Set** grade score s_i based on the grade of x_i in the golden distribution for all $x_i \in D$
4: **while** not reach termination criteria **do**.
5: **Update** θ_S such that $\sum_{x_i, x_j \in D} L_{Total}(x_i, x_j)$ is minimized, where θ_S is the parameter set of Siamese neural network.
6: **end while**

4. Experiment

4.1. Data overview

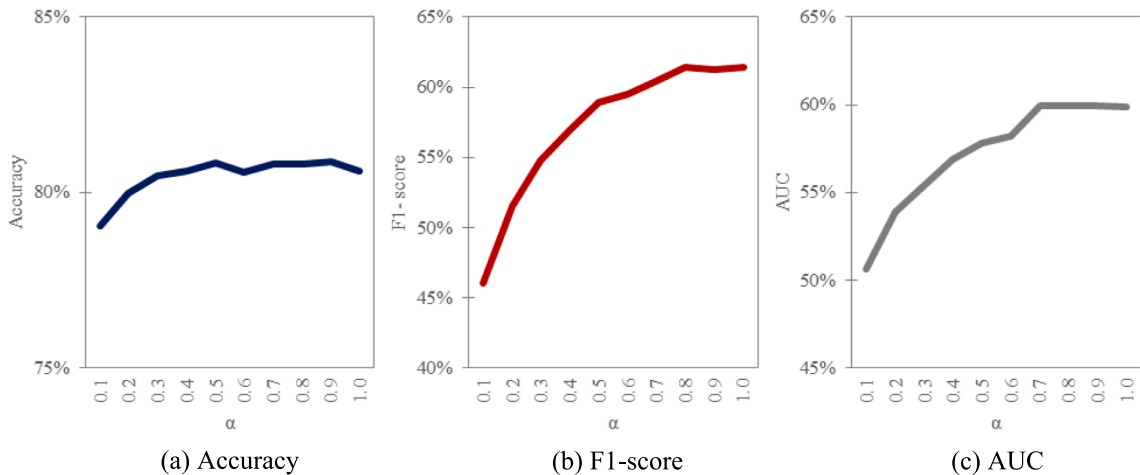
This study utilizes data from the Lending Club, an online P2P lending company in the United States. The Lending Club data include approximately 2.26 million P2P loan transaction records from June 2007 to December 2018. The data comprise 148 variables that capture various customer attributes, including personal information, application information, debt information, transaction history, and credit information, along with loan status for predicting customer defaults. The dataset is available at <https://www.kaggle.com/datasets/wordsforthewise/lending-club>.

The loan status indicates the debt repayment condition after loan execution and is categorized into six attributes: "Fully-paid," "Charge-off," "Current," "In grace period," two types of delinquency periods ("16–30 Days" and "31–120 days"), and "Default." This study predicts the default of borrowers whose repayment periods have expired. Thus, we only include customers whose loan status is either "Fully-paid" or "Charge-off," defining them as Non-default ($y = 0$) and Default ($y = 1$), respectively. As shown in Figs. 6, P2P lending data are imbalanced because Non-default and Default are unequal (Chen et al., 2021).

4.2. Experimental setup

4.2.1. Data preprocessing

We apply a data preprocessing method specifically designed for

**Fig. 7.** Average Accuracy, F1-score and AUC depending on the value of α .

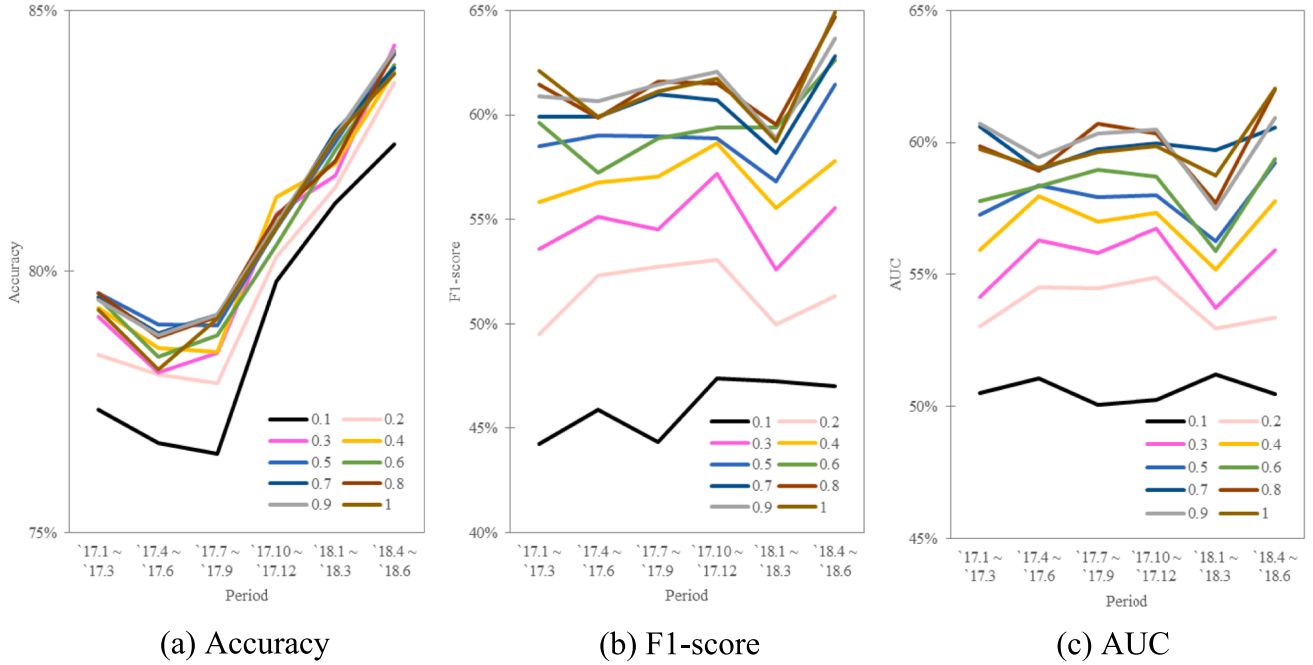


Fig. 8. Accuracy, F1-score, and AUC results by test period, based on the value of α .

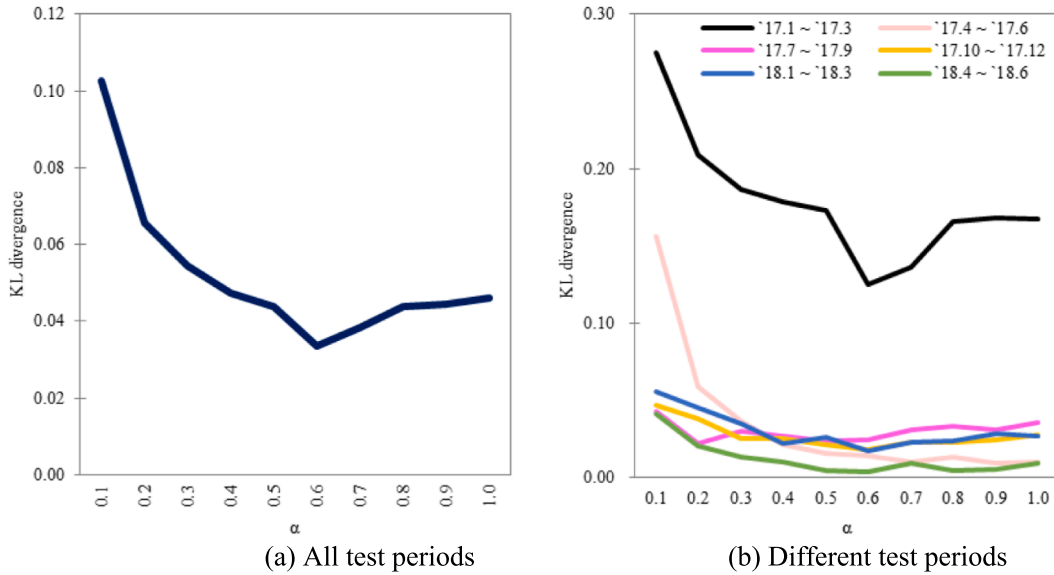


Fig. 9. Average KL divergence against the value of α .

Lending Club data, as proposed by [Vinod Kumar et al. \(2016\)](#). In our study, variables irrelevant to debt prediction, such as ID and zip codes; information only available after loan execution, such as collection recovery fees; and variables with a 35 % missing rate or higher are excluded. Some key variables, such as the total installment account balance and number of accounts opened in the last 6 months, were collected only after December 2015. Therefore, we use data from 2016 to 2018, which contain 517,051 transaction records. All variable values are normalized to a $[0, 1]$ range using min-max normalization by removing 1 % of the outliers. This is because when handling outliers in credit scoring model development, it is not advisable to remove a wide range of outliers ([Kim & Cho, 2019](#)). First, outliers can be considered non-default or default customers in credit scoring. When developing credit scoring models, a financial company must include default

customers with high loss potential. Therefore, the scope of outliers cannot be set too wide. Second, a financial company must calculate and produce credit scoring results for customers who apply for loans. Excluding too many outliers may lead to inconsistencies between the development and operation samples, resulting in distorted results. After data preprocessing, 78 variables are used in the model. These variables can be divided into the following five main categories([Aleksandrova, 2021](#)):

- General profile of the borrower – employment length, home ownership;
- Credit profile of the borrower – annual income, earliest credit line, FICO score, and others;
- Loan characteristics – purpose, term, funded amount;

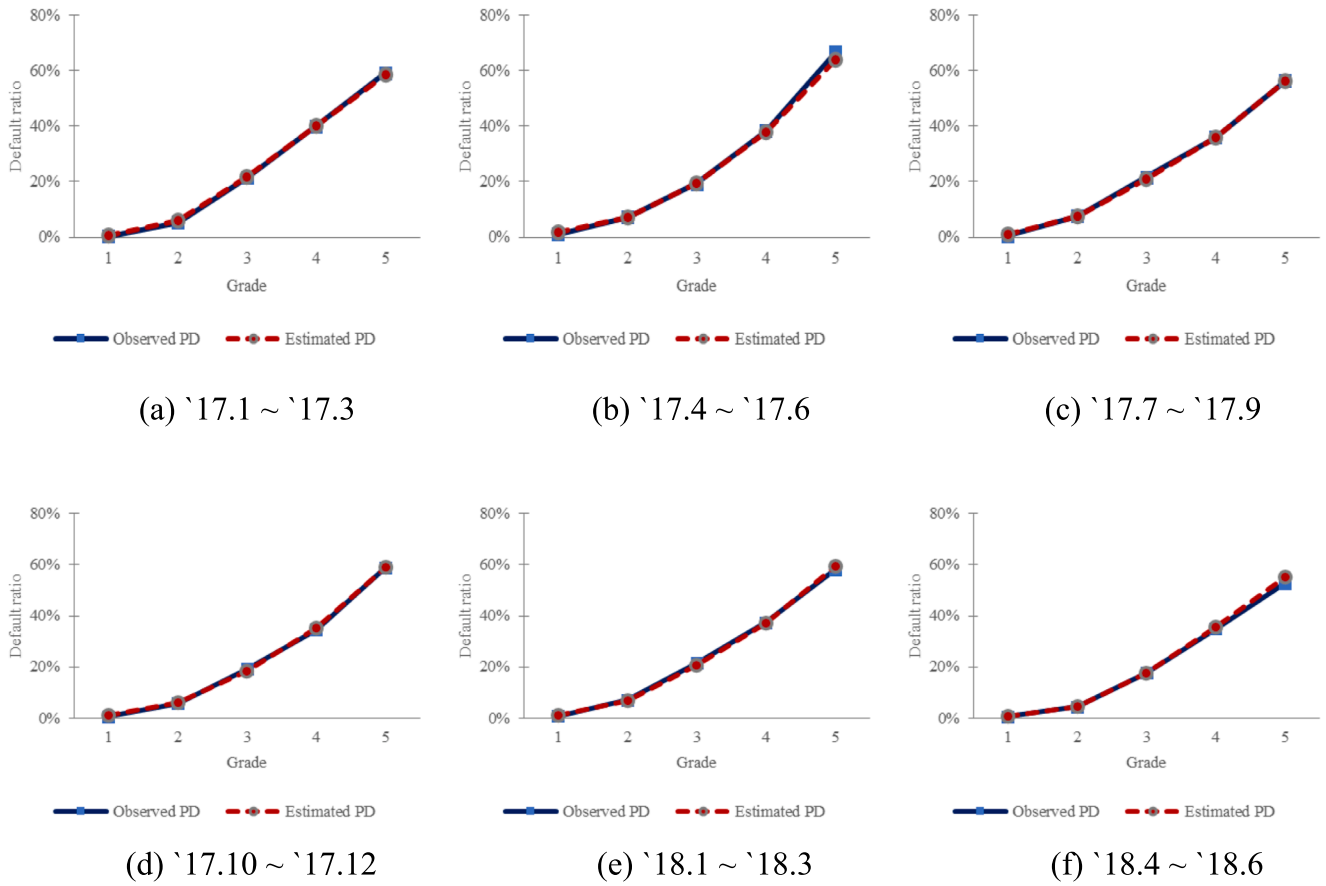


Fig. 10. Calibration results of proposed model by test period.

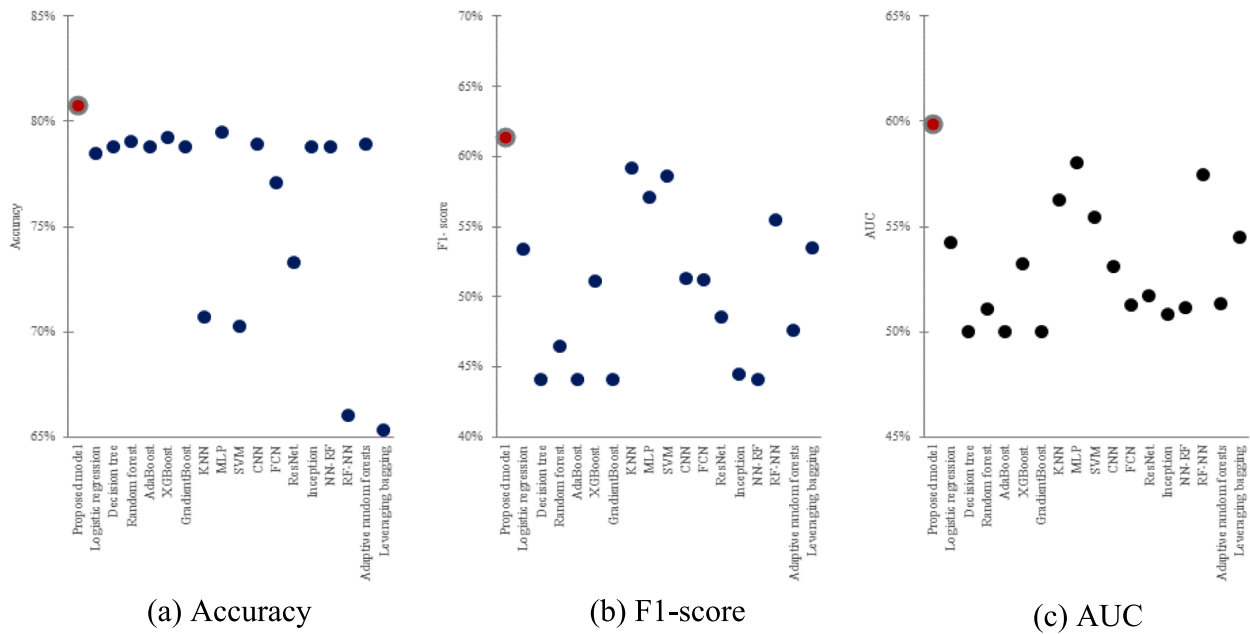


Fig. 11. Average Accuracy, F1-score and AUC results between the proposed model and comparison models.

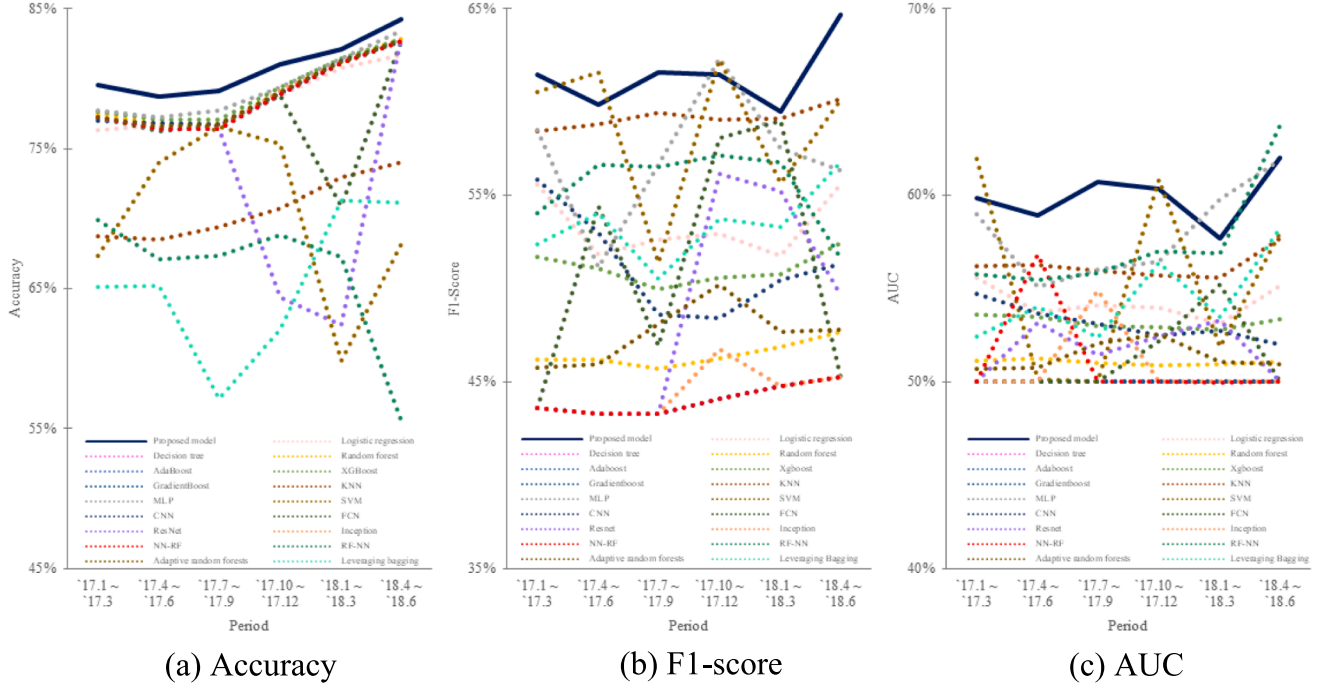


Fig. 12. Accuracy, F1-score and AUC results by test period between the proposed model and comparison models.

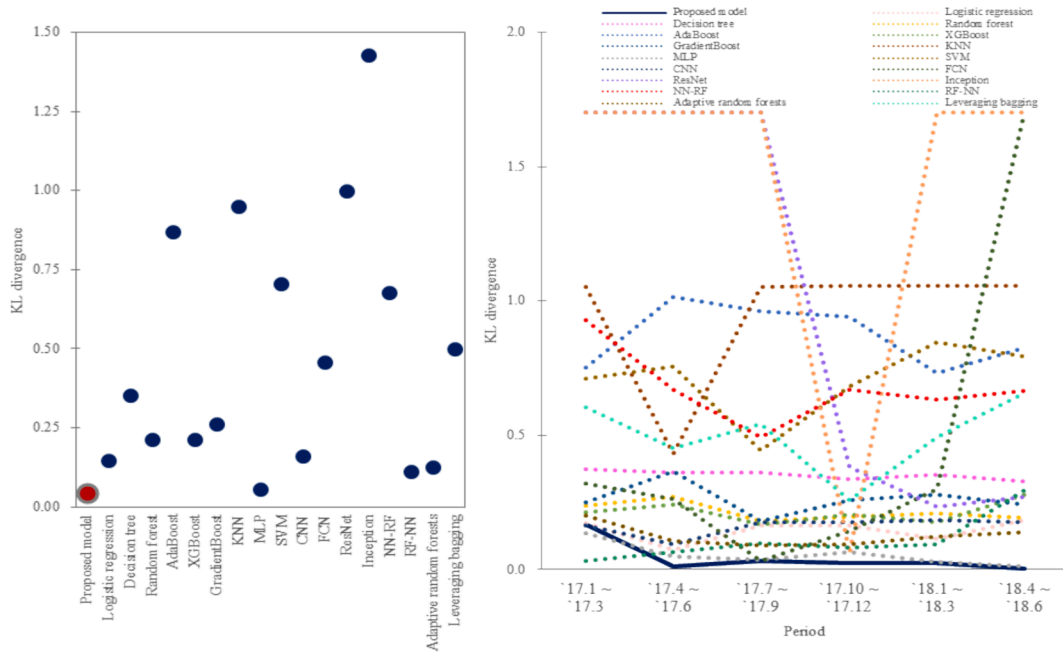


Fig. 13. Average KL divergence results for the proposed model and comparison models.

- Indebtedness – debt to income ratios, credit utilization variables and others.

Categorical variables are converted into dummy variables. Some minority categories are merged into one category (Wang et al., 2018).

4.2.2. Implementation details

Our proposed approach uses a DNN to generate the probability of default, followed by the generation of a target golden distribution and the application of a Siamese neural network to secure both predictive

power and stability in credit scoring. Both the DNN for creating the golden distribution and the Siamese network consist of three hidden layers with the rectified linear unit (ReLU) activation function; the number of nodes in each layer is defined as [78, 38, 19], given that there are 78 variables. We use the same number of hidden layers and apply the ReLU activation function, following Bastani et al. (2019). The learning rate is set to 0.001, and the Adam optimizer is used for training, with a batch size of 100.

We conduct two experiments to verify the performance of the proposed model. In summary, each experiment is conducted during the

training and test periods as outlined in Table 1. The first experiment validates the performance of the proposed model over a long period in which the default rate does not change significantly. In Experiment 1, we apply the moving window method to verify model performance over time. We train the model using the most recent one-year data based on the relevant quarter and shift the test period to three-month intervals. We set the balancing parameter α that determines the ratio of predictive power to stability as 0.8. Changes in predictive power and stability based on α are further detailed in Section 4.3.1.2 Sensitivity Analysis.

We find that when α is 0.8 or higher, the model's performance in terms of predictive power and stability is not significantly impacted by α .

The second experiment aims to validate the predictive power and stability of the model in an environment where the default rate rapidly changes over a short period. Therefore, we select time points where there is a significant difference in the default rates between the training and testing periods. Period 1 is the experimental environment where the default rate sharply decreases, whereas Period 2 is the experimental setting where the default rate sharply increases. We set the α to 0.7, as

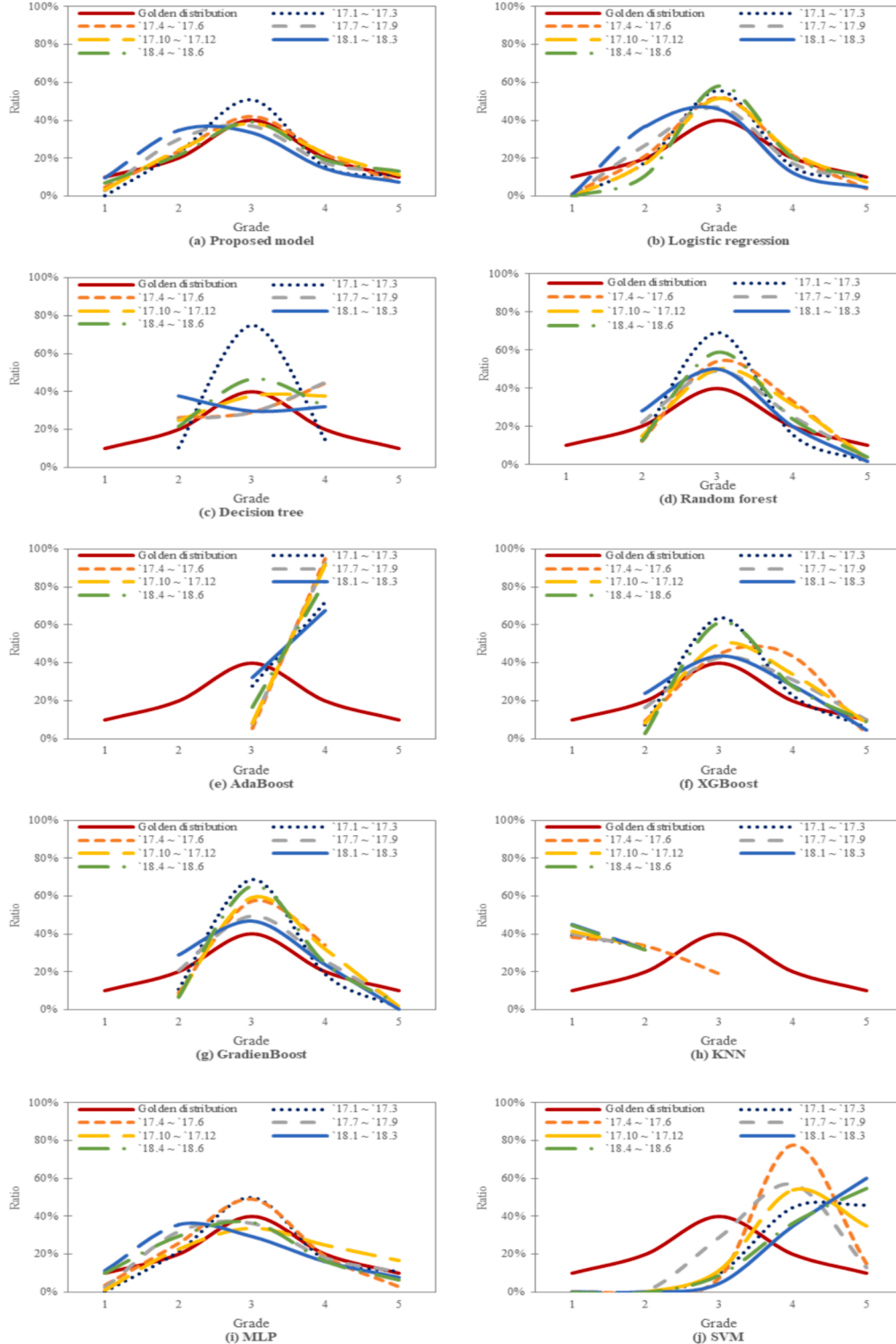


Fig. 14. Credit grade distribution across test periods for the proposed model and comparison models.

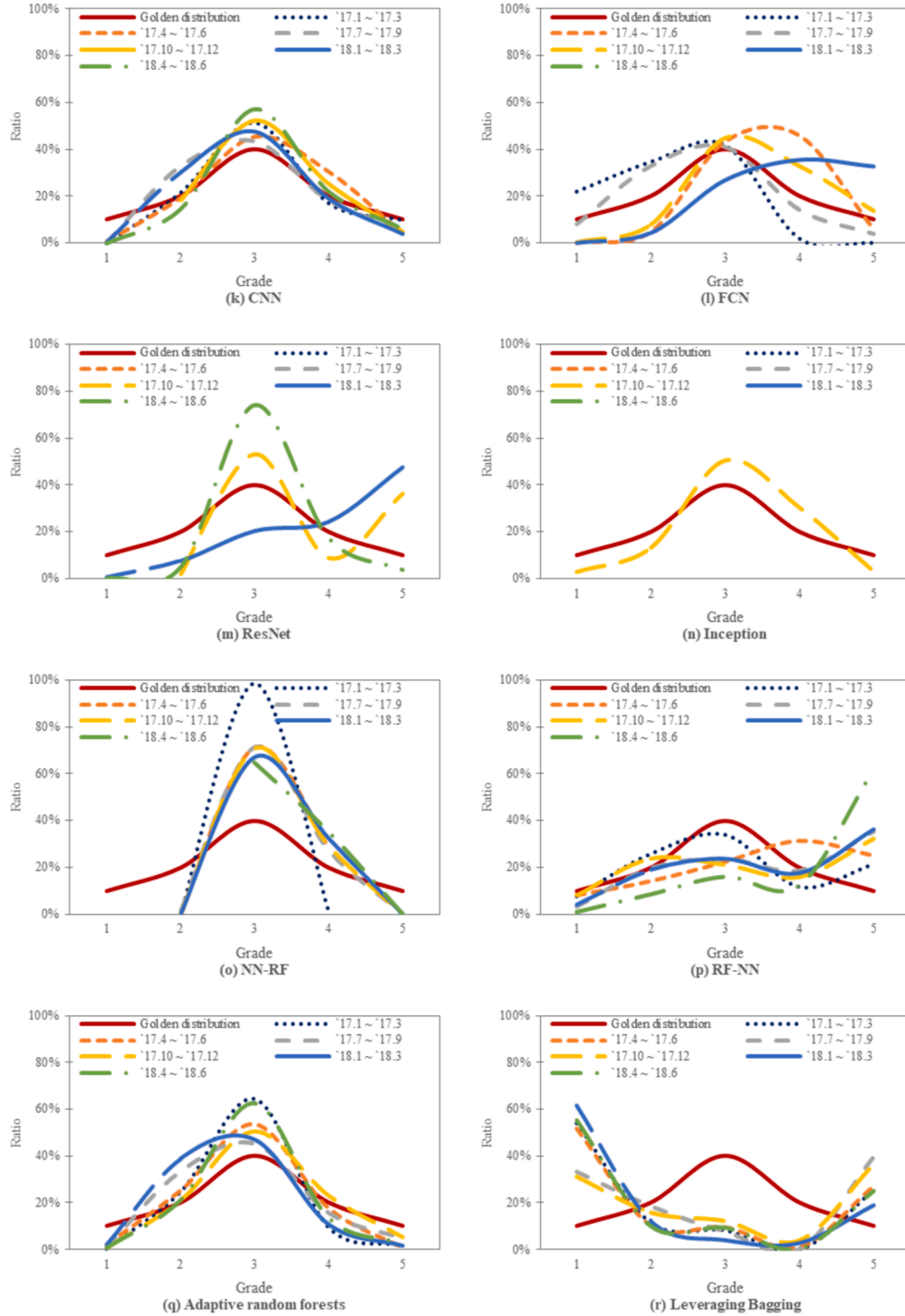


Fig. 14. (continued).

detailed in Section 4.3.2.2. When α is 0.7 or higher, the model's predictive power and stability is not significantly affected by α .

We conduct the experiments using the Python programming language on a personal computer equipped with an Intel Core i5 processor running at 1.8 GHz, 8.0 GB of RAM, and an Intel UHD Graphics 620. The TensorFlow framework is used to train the proposed model.

4.2.3. Comparison models

In this study, we assess the performance of our proposed model by comparing it with shallow machine learning models, including logistic regression, XGBoost, and random forest, as well as state-of-the-art deep

learning models (see Table 2). The deep learning models include CNN, multilayer perceptron (MLP) (Wang et al., 2016), fully convolutional networks (FCN) (Wang et al., 2016), ResNet (He et al., 2016), and Inception (Szegedy et al., 2014). Although CNN-based deep learning models generally use two-dimensional array-like data such as images, we utilize a CNN model based on 1-dimensional convolution layers because the Lending Club data are one-dimensional time-series data (Ismail Fawaz et al., 2019). Additionally, for comparison, we include models specifically designed for Lending Club data. Specifically, Fu (2017) proposes two credit scoring models. The first model calculates the probability of default for input samples using neural networks and

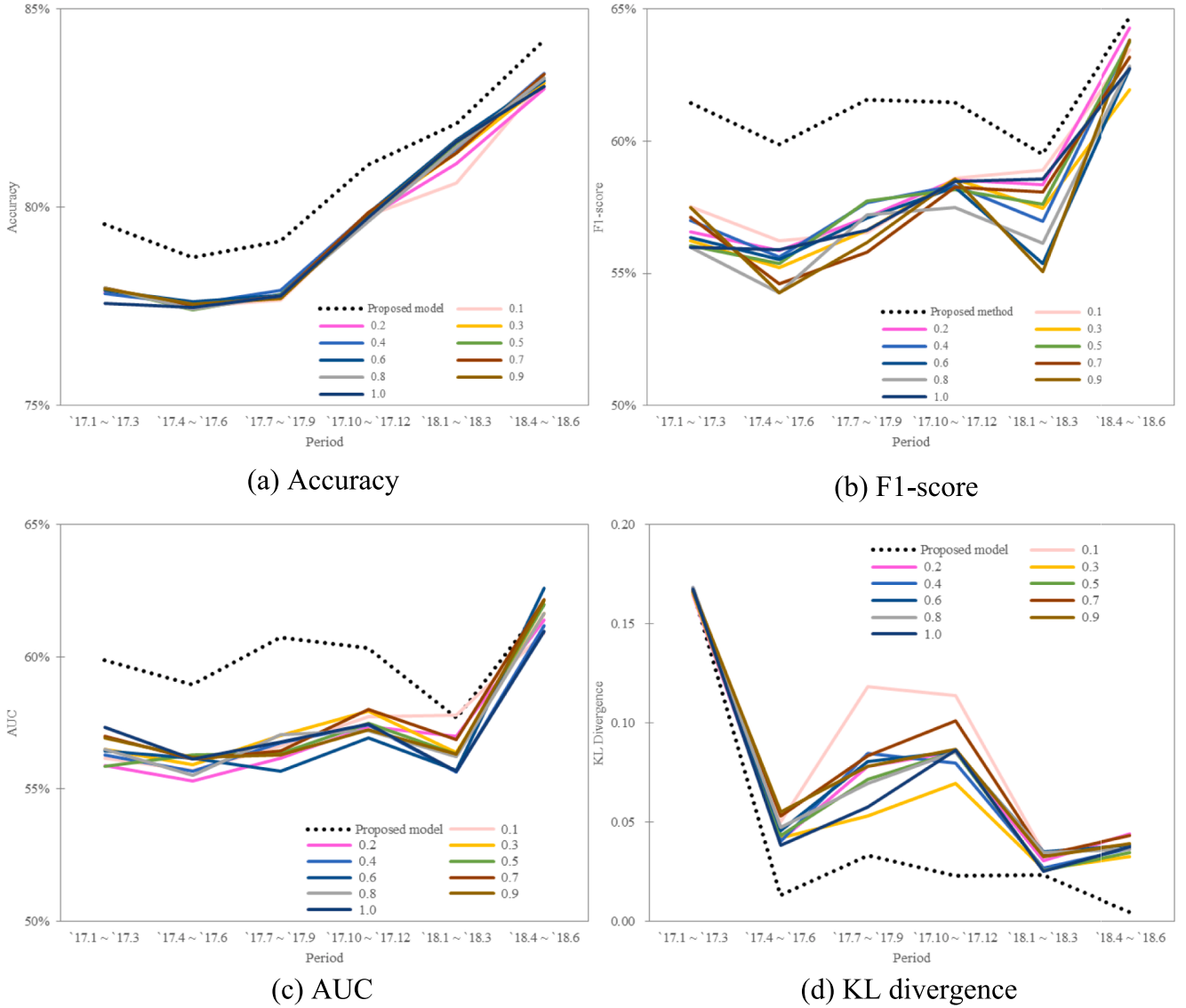


Fig. 15. Accuracy, F1-score, AUC and KL divergence results for the proposed model and multi-task NN model.

Table 6

Results of the golden distribution from P2P Lending Club data for a short-term period.

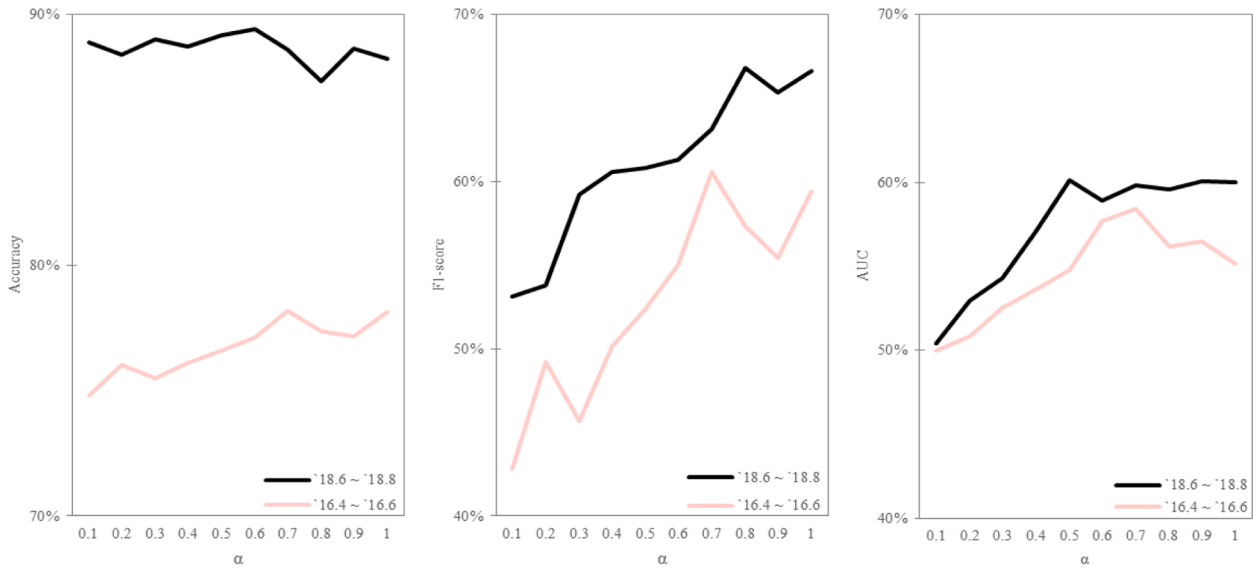
Credit Grade	'17.9 ~ '18.5			'16.1 ~ '16.3		
	Number of customers	Composition ratio	Score	Number of customers	Composition ratio	Score
Grade 1	8,305	10.0 %	0.00	11,179	10.0 %	0.00
Grade 2	16,610	20.0 %	0.06	22,357	20.0 %	0.05
Grade 3	33,221	40.0 %	0.26	44,713	40.0 %	0.28
Grade 4	16,610	20.0 %	0.49	22,357	20.0 %	0.50
Grade 5	8,305	10.0 %	0.92	11,179	10.0 %	0.97

uses it as an input for random forest to classify customer default. The second model uses random forest to calculate the probability of default and then uses it as an input for a neural network to classify customer defaults. In this study, we refer to Fu's first model as NN-RF and the second as RF-NN. We also include the data stream classification models adaptive random forest and leveraging bagging (Barddal et al., 2020) among the comparison models, which adapt their model parameters according to the changing data distribution over time. We set the parameters primarily based on the default values in the scikit-learn package, which is widely used for training machine learning algorithms,

including comparison models (see Table 3).

Finally, we include a customized neural network model as a comparison model, which introduces multi-task learning to simultaneously secure both predictive power and stability. We refer to this model as multi-task NN. Multi-task NN is defined with the same structure as the DNN for the golden distribution. In this model, we introduce multitask learning to further consider stability. The loss function for multi-task NN is defined as

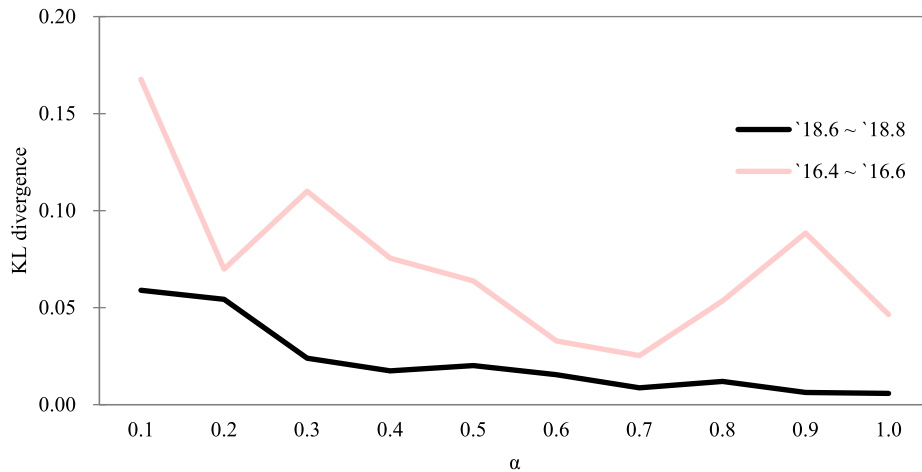
$$L_{Total}(x_i) = \alpha L_{BCE}(x_i) + (1 - \alpha)(\hat{y}_i - s_i)^2 \quad (6)$$



(a) Accuracy

(b) F1-score

(c) AUC

Fig. 16. Accuracy, F1-score and AUC results by each test period, based on the value of α .Fig. 17. Average KL divergence against the value of α .

where we vary the value of α between 0.1 and 1 in increments of 0.1 in a manner similar to the procedure described in Section 4.3.1.2. The equation for L_{MSE} can be found in Equation (4). The multitask NN model is included as a comparison model to assess the contribution of contrastive learning utilized in the proposed model (see Sections 4.3.1.5 and 4.3.2.5 for details).

4.2.4. Performance metrics

We verify both the predictive power for default risk and the stability of credit score distributions. The predictive power of credit scoring models is measured using various metrics (Řezáč, 2011); this study evaluates predictive power using Accuracy and F1-score. Let TP be the number of correctly predicted non-default customers, FN the number of incorrectly predicted non-default customers, FP the number of incorrectly predicted default customers, and TN the number of correctly predicted default customers. Accuracy and F1-score are defined in Equations (7) and (8), respectively, where $Precision = \frac{TP}{TP+FP}$, $Recall = \frac{TP}{TP+FN}$. The cutoff value to differentiate between classes is 0.5 for the F1-

score.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

$$F1 - score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (8)$$

The area under the curve (AUC) is also used to evaluate prediction performance for imbalanced data in credit scoring. The AUC measures the area under the receiver operating characteristic curve, wherein the x-axis is defined as False positive rate and y-axis as True positive rate, where False positive rate = $\frac{FP}{FP+TN}$ and True positive rate = $\frac{TP}{TP+FN}$ (Forman & Cohen, 2005).

We also evaluate whether the distribution of the default probability (credit score) generated by the proposed model remains stable over time as new samples are collected. To this end, we calculate the difference between the probability distribution for new samples (P) generated by the proposed model and the initially defined golden distribution (Q) using Kullback-Leibler (KL) divergence in Equation (9). Here, p_i

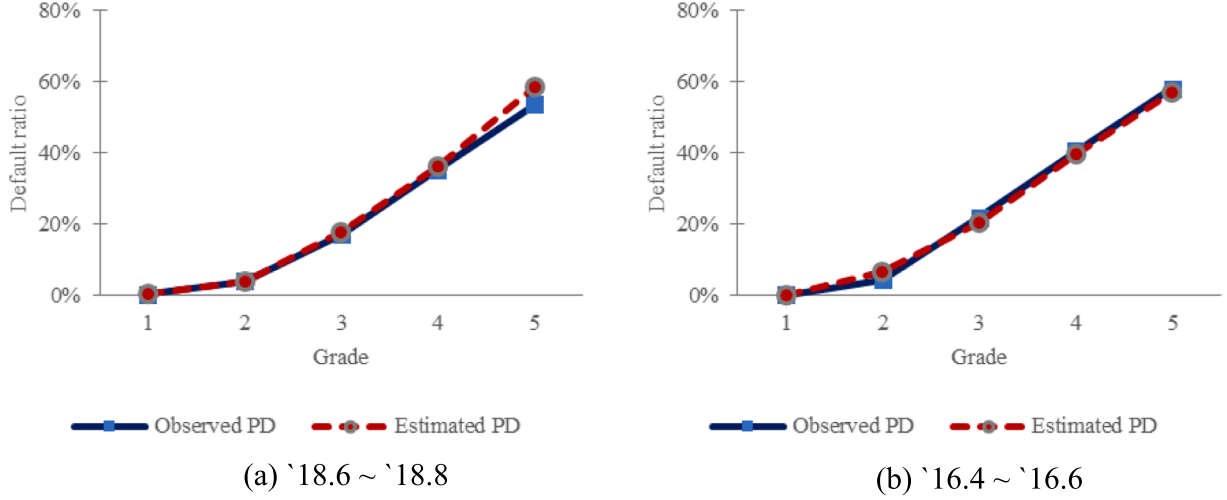


Fig. 18. Calibration results of proposed model by test period.

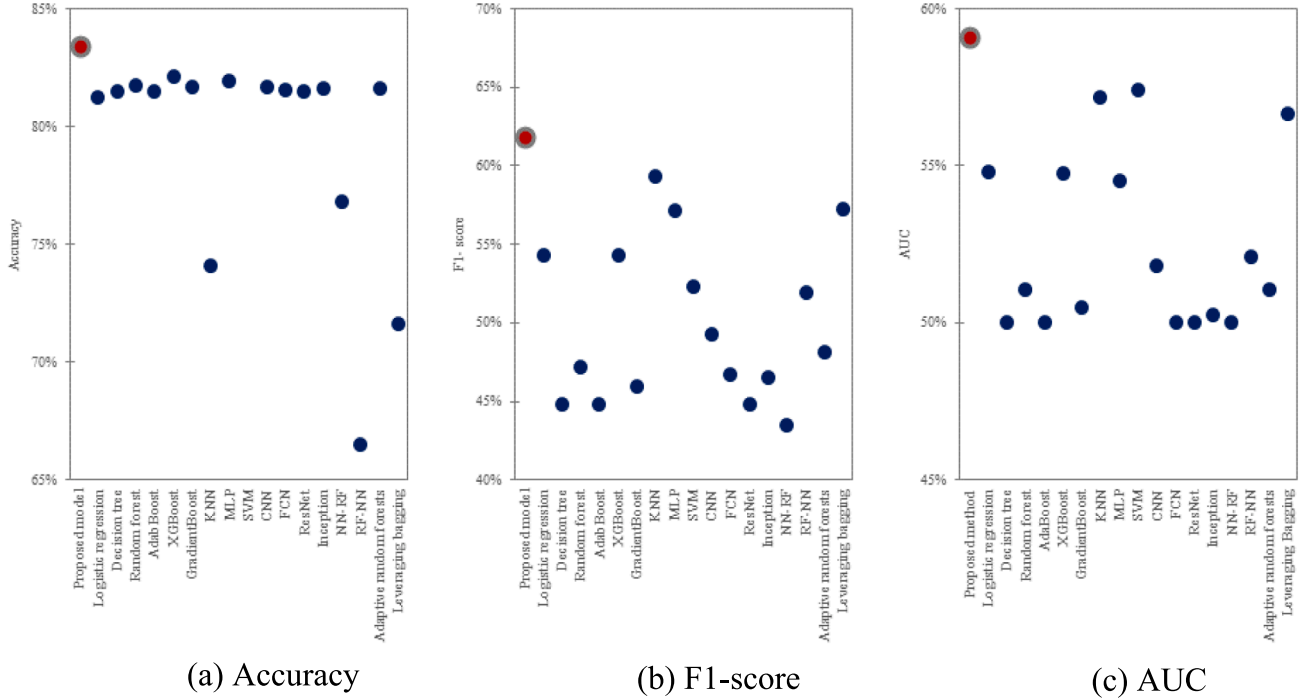


Fig. 19. Average Accuracy, F1-score, and AUC results between the proposed model and comparison models.

represents the composition ratio of the i th credit grade as determined by the proposed model, with credit grade intervals defined in the same manner as those defined in the golden distribution. p_i can be obtained by checking whether the default probability of each sample calculated by the DNN falls within the predefined default probability interval of the i th grade of the golden distribution. q_i refers to the composition ratio of the i th grade in the predefined golden distribution. q_i can be obtained by determining the default probability threshold that satisfies the composition ratio of the predefined i th grade as described in Section 3.1. M is the number of credit grade bins, which we define as 5 in this study. However, in some instances, certain credit grade bins in the distribution generated by the proposed model could have a composition ratio of zero; in such cases, we replace $\frac{p_i}{q_i}$ with q_i for the calculation (Boyd & Vandenberghe, 2004).

$$D_{KL}(P||Q) = \sum_{i=1}^M p_i \log\left(\frac{p_i}{q_i}\right) \quad (9)$$

4.3. Experiment results

4.3.1. Experiment 1: Long-term period

4.3.1.1. Golden distribution generation. The golden distribution is the target distribution of scores generated using the proposed model. After computing the default probabilities of customers using a DNN, we categorize them into the 5-grade system. The results are presented in Table 4. As shown in Fig. 6, as the default probabilities of customers change according to the loan origination period, the credit score values for each credit grade vary over different learning periods. We compare

Table 7

Accuracy and F1-score results by test period between the proposed model and comparison models.

Models	Accuracy		F1-score		AUC	
	'18.6 ~ '18.8	'16.4 ~ '16.6	'18.6 ~ '18.8	'16.4 ~ '16.6	'18.6 ~ '18.8	'16.4 ~ '16.6
Proposed model	88.6 %	78.2 %	63.2 %	60.6 %	59.8 %	58.4 %
Logistic regression	87.1 %	75.5 %	57.1 %	51.5 %	55.8 %	53.8 %
Decision tree	88.2 %	74.8 %	46.9 %	42.8 %	50.0 %	50.0 %
Random forest	88.4 %	75.1 %	49.6 %	44.8 %	51.3 %	50.8 %
AdaBoost	88.2 %	74.8 %	46.9 %	42.8 %	50.0 %	50.0 %
XGBoost	88.3 %	76.0 %	56.5 %	52.1 %	55.2 %	54.3 %
GradientBoost	88.2 %	75.1 %	46.9 %	45.0 %	50.0 %	50.9 %
KNN	78.1 %	70.0 %	59.0 %	59.7 %	57.6 %	56.7 %
MLP	87.6 %	76.2 %	61.9 %	52.4 %	54.6 %	54.5 %
SVM	67.8 %	48.3 %	56.4 %	48.3 %	57.5 %	57.3 %
CNN	88.1 %	75.3 %	50.0 %	48.6 %	51.3 %	52.3 %
FCN	88.3 %	74.8 %	50.3 %	43.2 %	50.0 %	50.0 %
ResNet	88.2 %	74.8 %	46.9 %	42.8 %	50.0 %	50.0 %
Inception	88.2 %	75.0 %	46.9 %	46.1 %	50.0 %	50.5 %
NN-RF	77.2 %	76.4 %	43.6 %	43.3 %	50.0 %	50.0 %
RF-NN	58.7 %	74.4 %	52.9 %	51.0 %	53.0 %	51.2 %
Adaptive random forests	88.0 %	75.2 %	49.6 %	46.8 %	50.7 %	51.5 %
Leveraging bagging	77.0 %	66.2 %	57.0 %	57.6 %	56.8 %	56.4 %

the average composition ratios per credit grade between the training and test periods to verify that the golden distribution estimated by the DNN model has sufficient predictive power. Table 5 summarizes the results. Evidently, the distribution of the predefined composition ratios in the training period and the composition ratios produced by the DNN during the test period are similar. In other words, the DNN for the golden distribution has sufficient accuracy for successive model training.

4.3.1.2. Sensitivity analysis. In this section, we conduct a sensitivity analysis to assess the impact of varying the balancing parameter α of the proposed multi-task learning model on its predictive power and stability. For the sensitivity analysis, we vary the value of α between 0.1 and 1 in increments of 0.1 and measure the changes in Accuracy, F1-score, AUC, and KL divergence. Fig. 7 presents the average Accuracy, F1-score, and AUC across six test intervals, based on the value of α . The results reveal that as the value of α increases, the importance of prediction loss increases, leading to an improvement in Accuracy, F1-score, and AUC. Specifically, Accuracy improves from 79.0 % to 80.9 % as the value of α increases. However, Accuracy remains around 80.8 % when the value of α is at least 0.5. On the other hand, F1-score shows a substantial improvement, increasing from 46.0 % to 61.4 % as the value of α

increases. However, F1-score plateaus at around 61.4 % when the value of α is at least 0.8. The AUC increases from 50.6 to 59.9 as α increases. However, the AUC does not increase above an α value of 0.7, which is similar to the F1-score result.

Fig. 8. shows the predictive power across the different test periods. Overall, as it approaches more recent time points, the Accuracy, F1-score, and AUC metrics increase, indicating enhanced predictive power. More importantly, the trend of the predictive power remains similar across various α . This implies that we can determine α based on the mean predictive power, as shown in Fig. 7.

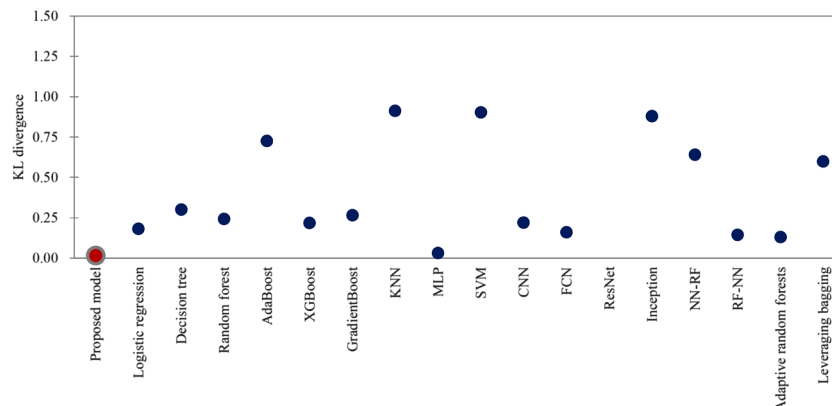
Stability is also evaluated in a manner similar to the predictive sensitivity analysis. We first examine the average stability across all test intervals and then examine the stability for individual intervals. Fig. 9(a) shows the average changes in KL divergence across all test intervals according to the value of α . Specifically, KL divergence significantly decreases within the range of 0.1–0.2 for α . The lowest KL divergence is found at $\alpha = 0.6$. This is because, as the value of α increases, a certain level of predictive power is ensured, preventing the scores from concentrating in a certain grade. Consequently, the distribution of the generated scores become similar to the golden distribution. However, when the value of α exceeds 0.6, the KL divergence value increases. This is because, as the value of α increases above $\alpha = 0.6$, the importance of stability loss decreases.

Fig. 9(b) displays the changes in stability by test intervals depending on the values of α . Generally, the KL divergence value decreases as we move closer to more recent lending periods. Particularly in the test period from '17.1 to '17.3, the KL divergence value is high because the score proportion of grade 1 is calculated to be around 0.1 %, regardless of the value of α . However, over time, as the grade 1 composition ratio increases to over 3 %, and the composition ratios of grades 2–4

Table 8

KL divergence results by test period between the proposed model and comparison models.

Models	'18.6 ~ '18.8	'16.4 ~ '16.6	Models	'18.6 ~ '18.8	'16.4 ~ '16.6
Proposed model	0.01	0.03	SVM	0.73	1.08
Logistic regression	0.19	0.18	CNN	0.25	0.19
Decision tree	0.33	0.28	FCN	0.14	0.18
Random forest	0.25	0.23	ResNet	1.70	1.70
AdaBoost	0.75	0.70	Inception	1.70	0.06
XGBoost	0.24	0.20	NN-RF	0.69	0.59
GradientBoost	0.33	0.21	RF-NN	0.20	0.09
KNN	1.08	0.75	Adaptive random forests	0.10	0.16
MLP	0.02	0.04	Leveraging bagging	0.49	0.71

**Fig. 20.** Average KL divergence results for the proposed model and comparison models.

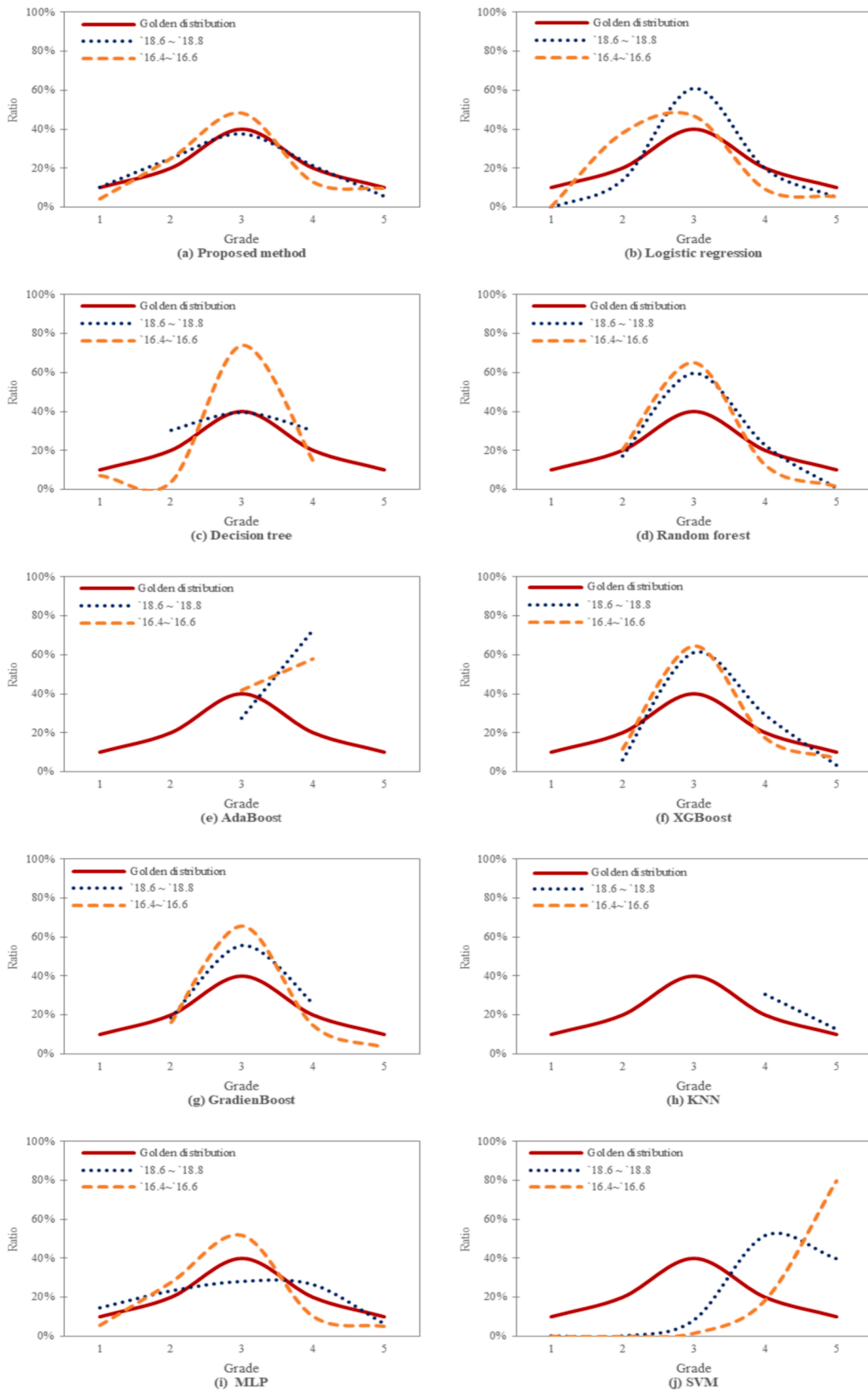


Fig. 21. Credit grade distribution across test periods for the proposed model and comparison models.

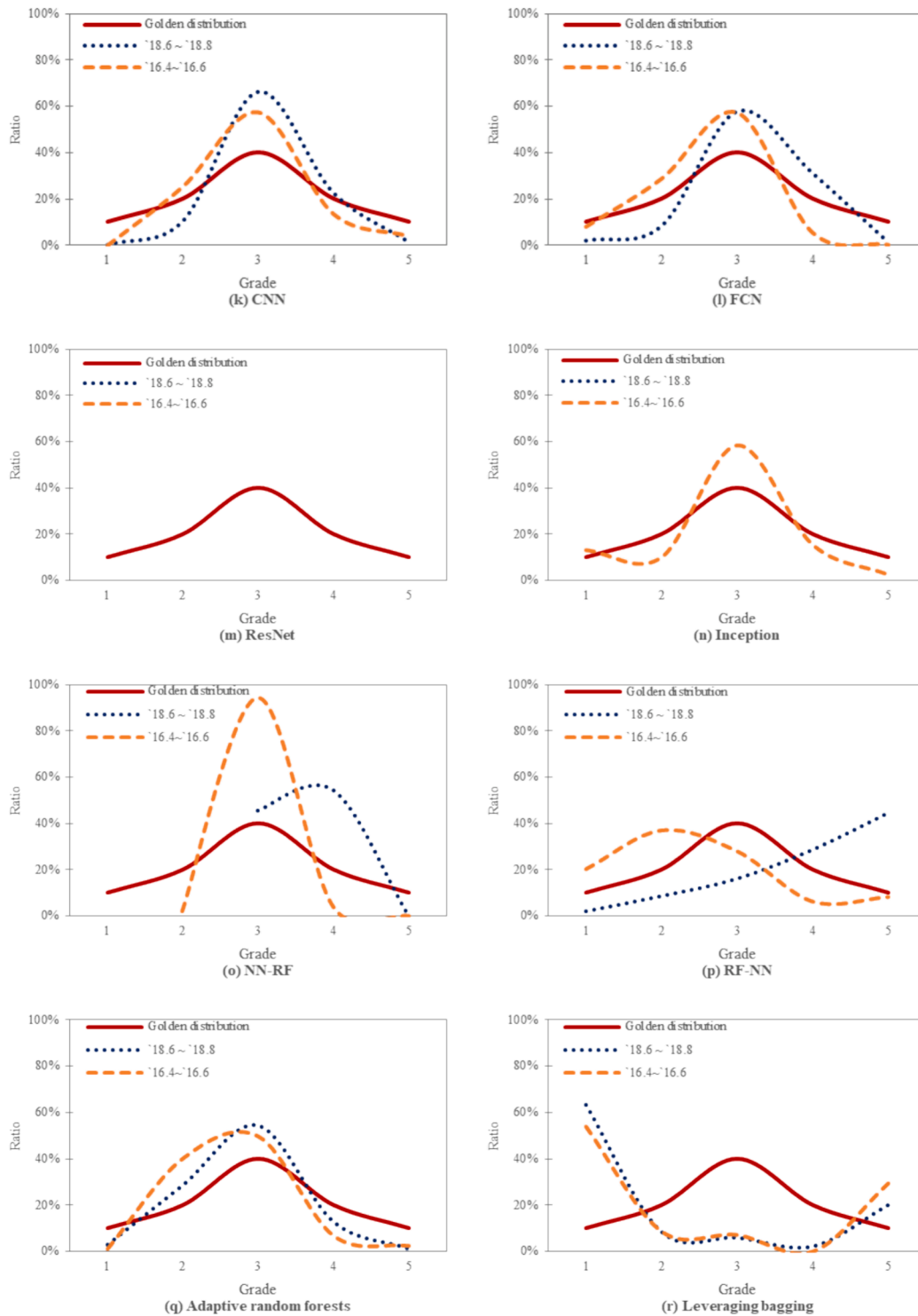


Fig. 21. (continued).

resembles the golden distribution, the overall KL divergence value declines. Considering both prediction power and stability, we recommend α between 0.8 and 0.9, and set α as 0.8.

From the sensitivity analysis results, we can observe that the proposed model consistently maintains its predictive power without periodic retraining. This is because the alpha value can be set to ensure high predictive accuracy across different time periods through validation. Furthermore, our proposed model ensures the robustness of the prediction by considering stability during training.

Additionally, we observe the calibration of the model between the

observed and estimated PD. Calibration is a technique for mapping the credit score to the credit grade and then estimating the PD per credit grade (Burgt, 2019). It is a dimension of scorecard quality and a regulatory requirement (Bequé et al., 2017). To verify the model calibration, we compare the default ratio by grade between the observed and estimated PD for each test period. As shown in Fig. 10, the default ratio trend by grade is very similar between the observed and estimated samples for each period. That is, the proposed model is well-calibrated in Experiment 1.

Table 9

Accuracy, F1-score, and KL divergence results for the proposed model and multi-task NN model.

		Accuracy		F1-score		AUC		KL divergence	
		'18.6 ~ '18.8	'16.4 ~ '16.6	'18.6 ~ '18.8	'16.4 ~ '16.6	'18.6 ~ '18.8	'16.4 ~ '16.6	'18.6 ~ '18.8	'16.4 ~ '16.6
Proposed model		88.6 %	78.2 %	63.2 %	60.6 %	59.8 %	58.4 %	0.01	0.03
	$\alpha = 0.1$	87.8 %	75.8 %	49.0 %	60.9 %	57.8 %	54.2 %	0.13	0.15
	$\alpha = 0.2$	88.2 %	75.6 %	47.5 %	60.6 %	59.5 %	53.1 %	0.17	0.18
	$\alpha = 0.3$	87.5 %	75.5 %	47.6 %	64.2 %	58.3 %	53.5 %	0.17	0.13
	$\alpha = 0.4$	87.5 %	75.8 %	49.0 %	64.0 %	58.9 %	52.4 %	0.10	0.13
Muli-task NN	$\alpha = 0.5$	87.8 %	75.3 %	45.8 %	61.2 %	58.5 %	53.6 %	0.08	0.12
	$\alpha = 0.6$	87.5 %	75.8 %	49.2 %	62.2 %	57.7 %	53.7 %	0.08	0.12
	$\alpha = 0.7$	87.1 %	76.3 %	53.0 %	65.4 %	59.5 %	53.5 %	0.07	0.13
	$\alpha = 0.8$	88.0 %	76.3 %	54.0 %	61.9 %	60.1 %	54.4 %	0.05	0.10
	$\alpha = 0.9$	87.9 %	75.8 %	49.3 %	62.8 %	57.9 %	52.4 %	0.07	0.12
	$\alpha = 1.0$	87.5 %	76.3 %	54.3 %	64.5 %	59.4 %	54.6 %	0.04	0.12

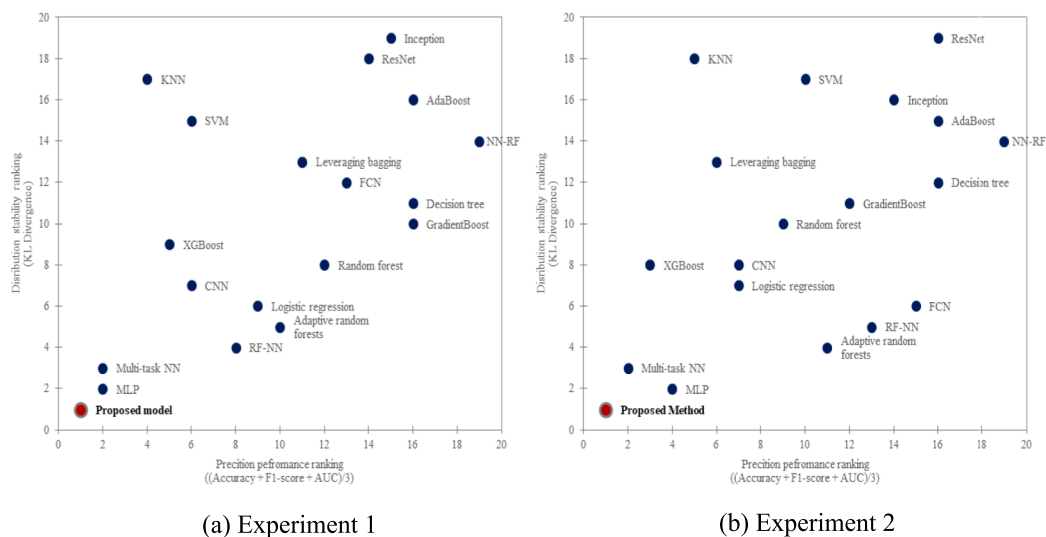


Fig. 22. Comparison of prediction performance and stability rankings.

4.3.1.3. Prediction performance comparison. This section compares the predictive power of the proposed model and the existing credit scoring models. First, we measure the Accuracy, F1-score, and AUC for various test periods using both the proposed and comparison models, with the comparison based on the average performance across all test periods (see Fig. 11). The proposed model consistently outperforms all comparison models in terms of average predictive power. Specifically, the proposed model’s Accuracy, F1-score, and AUC results are superior even when compared with state-of-the-art deep learning models such as CNN, MLP, FCN, ResNet, and Inception, as well as models specialized in improving credit scoring prediction performance, such as NN-RF and RF-NN. We originally introduce contrastive learning to improve the model stability. However, the experimental results demonstrate that contrastive learning improves the predictive power of the proposed model by increasing its robustness. In other words, we do not find a complete tradeoff between the prediction loss and difference loss because the difference loss acts as a regularizer.

We also compare the Accuracy, F1-score, and AUC results for the individual test periods. The results are summarized in Fig. 12. According to the experiments, the proposed model exhibits superior predictive power during most test periods. Unlike some models such as SVM and ResNet, which show significant fluctuations in performance across different periods, the proposed model consistently maintains high performance, ultimately securing a high average Accuracy, F1-score and AUC compared with the other models. Moreover, FCN, Inception, and NN-RF show increasing accuracy as they approach more recent data points, although their F1-scores remains stable. This is because these

models predominantly classify predictions as non-default, resulting in lower precision scores. From the perspective of actual financial institutions managing credit evaluation models, this result suggests that the proposed model can effectively minimize financial losses by accurately classifying default customers. In addition, Fig. 12(c) demonstrates that the proposed model is effective even in an imbalanced dataset environment, which is common in credit scoring (Namvar et al., 2018).

4.3.1.4. Stability comparison. This section comparatively analyzes the stability test results of the proposed and comparison models. Fig. 13(a) shows the average KL divergence across all the test periods for the proposed and comparison models. The proposed model exhibits the lowest average KL divergence value of 0.04. This indicates that the distribution of the scores generated by the proposed model consistently approximates the golden distribution.

Fig. 13(b) presents the stability evaluation results for the proposed and comparison models for each test period. Except for ‘17.1 ~ 17.3 period, the KL divergence value for the proposed model is the lowest in all test periods. This is due to the proposed model’s ability to adaptively learn the changing score distribution over time, particularly in a context in which default rates fluctuate with the loan execution time (see Fig. 6). Specifically, except for the ‘17.1-’17.3 period, the proposed model maintains a KL divergence value of 0.03 or lower, indicating a distribution very similar to the golden distribution.

Among the comparison models, SVM, FCN, RF-NN, and leveraging bagging fail to maintain a symmetrical bell-shaped probability distribution (see Fig. 14). Additionally, random forest, XGBoost, and

GradientBoost do not produce scores within the 1st grade, resulting in a 4-grade probability distribution. ResNet and Inception predict that all samples fall into a single grade in some test periods, and therefore do not generate a grade distribution for those periods (refer to Fig. 14(m) and (n)). In contrast, the proposed model maintains a 5-grade golden distribution across all test periods.

4.3.1.5. Comparison with multi-task NN. Fig. 15 shows the predictive power and stability for the different test periods. Overall, the proposed model outperforms multi-task NN model in terms of Accuracy, F1-score, AUC, and KL divergence, irrespective of the value of α . This result is meaningful because our contrastive learning with the multi-task learning concept is a better approach than multitask learning alone.

4.3.2. Experiment 2: Short-term period

4.3.2.1. Golden distribution generation. We generate the golden distribution using the method described in Section 4.3.1.1. As shown in Table 6, the credit score values for each credit grade differ for each learning period.

4.3.2.2. Sensitivity analysis. We also perform a sensitivity analysis, as described in Section 4.3.1.2. Fig. 16 shows the predictive power for each test period. In both test periods, Accuracy metric values are higher than F1-score regardless of α . While F1-score substantially increases compared with Accuracy, F1-score remains stable above $\alpha = 0.7$. The overall trend of the AUC metric is similar to that of the F1-score; however, the differences in AUC values across each test period are smaller than those of the F1-score. In each test period, the prediction power in '18.6 ~ '18.8, which rapidly declines in default, is higher than in '16.4 ~ '16.6.

Stability is also evaluated in a manner similar to that of the predictive sensitivity analysis. Fig. 17 represents average changes in KL divergence across each test intervals according to the value of α . '16.4 ~ '16.6 period has the higher KL divergence value than '18.6 ~ '18.8 period. It means that '16.4 ~ '16.6 period is a more difficult experimental setting than '18.6 ~ '18.8 period to achieve stability of the model. Considering both prediction accuracy and stability across the experiment periods, we recommend an α between 0.6 and 0.8, and set α at 0.7.

We also conduct experiments to verify the model calibration in Experiment 2. As shown in Fig. 18, the proposed model is well-calibrated in this setting as well.

4.3.2.3. Prediction performance comparison. In this section, we compare the predictive powers of the proposed and comparison models by measuring the average Accuracy, F1-score, and AUC. The proposed model has the highest average Accuracy, F1-score, and AUC among all the comparison models (see Fig. 19). Table 7 presents the Accuracy, F1-score, and AUC results for the individual test periods between the proposed and comparison models. In terms of Accuracy, decision tree and random forest only perform well in '18.6 ~ '18.8, while MLP and NN-RF perform well in '16.4 ~ '16.6. However, the proposed model outperforms all the comparison models in terms of both Accuracy and F1-score for the individual test periods. This means that the proposed model maintains high predictive power, even in an environment where default rates rapidly decrease or increase over a short period.

4.3.2.4. Stability comparison. This section presents the stability comparison results of the proposed model and the comparison models. As shown in Fig. 20, the proposed model has the lowest average KL divergence value of 0.02. For each test period, the proposed model has the lowest KL divergence (see Table 8). Fig. 21 shows the credit grade distribution across the test periods between the proposed model and the comparison models. The proposed model clearly maintains a 5-grade

golden distribution across both test periods. However, several comparison models, including Inception, XGBoost, and AdaBoost, even fail to maintain a 5-grade system.

4.3.2.5. Comparison with multi-task NN. For each test period, we also compare the proposed model with the multitask NN model to verify the contribution of the proposed contrastive learning. Table 9 presents the predictive power and stability results for each test period. In most test periods, the proposed model exhibits an overall better performance than the multitask NN model in terms of Accuracy, F1-score, AUC, and KL divergence. This result indicates that contrastive learning, as well as the multitask learning concept, is essential for short-term credit scoring.

4.3.3. Analysis of experiment results

Finally, we compare the predictive power and stability of the proposed model with those of the comparison models. Predictive power is calculated by averaging the Accuracy, F1-score, and AUC for each test period, whereas stability is calculated by averaging the KL divergence for each test period. These values are then ranked. The results of each experiment are shown in Fig. 22. The experimental results show that the proposed model performs the best in terms of both predictive power and stability. Specifically, the multi-task NN model exhibited the second-best performance in both experiments in terms of prediction power. The proposed model outperforms multi-task NN at $\alpha = 0.6$ by 1 % in terms of Accuracy, 7 % in terms of F1-score, and 5 % in terms of AUC in Experiment 1. In Experiment 2, the proposed model outperforms multi-task NN at $\alpha = 0.8$ by 2 % in terms of Accuracy, 7 % in terms of F1-score, and 2 % in terms of AUC in Experiment 2. When compared with the MLP, which has the second-best performance in both experiments in terms of KL divergence, the proposed model is superior by 25 % in Experiment 1 and by 47 % in Experiment 2.

5. Conclusion

This study introduces a multitask learning-based Siamese neural network and develops a credit scoring model that ensures both predictive power and stability. To ensure stability, we propose a new contrastive learning technique that leverages a golden distribution. We utilize P2P Lending Club data to construct an experimental environment similar to a real financial market and compare the performance of the proposed model with various machine and deep learning models. The proposed model yields superior predictive power for default compared with the comparison models and consistently maintains this high predictive power even when the loan execution time changes over the long term or when the default rate changes rapidly in the short term. In terms of stability, the proposed model maintains a probability distribution similar to the predefined 5-grade golden distribution. In conclusion, the proposed model demonstrates superior performance in terms of both predictive power and stability compared to the comparison models for various experimental settings.

In practice, financial companies suffer significant losses when their credit scoring models misclassify default customers as non-default. Therefore, it is crucial to define the cost of misclassifying default customers (Lohmann & Ohliger, 2019; Petrides et al., 2022) and minimize it to maximize financial companies' profits in the credit scoring field (Shen et al., 2020). Furthermore, as a topic for future research, we aim to develop new learning techniques to minimize potential losses owing to misclassification. We anticipate that this model will not only improve predictive power and stability, which are the core considerations of this study, but will also develop a new credit scoring model that can substantially minimize losses for financial companies.

CRedit authorship contribution statement

Soonjae Kwon: Conceptualization, Methodology, Software, Writing

– original draft. **Jaeyeon Jang:** Visualization, Validation, Writing – review & editing. **Chang Ouk Kim:** Project administration, Funding acquisition, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Chang Ouk Kim reports financial support was provided by National Research Foundation of Korea. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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